Vojtěch Kubáč

C++ engineer · CFD, FEA, applied ML

Experience

2024–today **Software engineer**, *Ententee*
Full-stack product delivery across healthcare and robotics engagements, developed alongside my role at CFD Support. Stack spans Python, Java / Spring Boot, Django, React / TypeScript, PostgreSQL, with ROS / Gazebo and C++ on the UAV side.
- **THK UAV** — greenfield stack: perception, mission autonomy, GPS-denied nav.
- **DrChrono** (2024–25) — US medical app uplift: Django/GraphQL, React/TS.
- **ALF** — medical-faculty app for attestations (Spring Boot, React/TS, PG).
- **Ententee Hub** — internal ops and hiring tooling (FastAPI, React/TS, PG).

2019–today **C++ software engineer**, *CFD Support s.r.o.*, Prague
Own multiple modules of TCAE, a cross-platform CAE desktop application built around ParaView/VTK and OpenFOAM. Core stack: C++23, CMake, Qt, GitLab CI.
- **TFEA** — FEA module: CalculiX, CFD-to-FEA mapping (FSI), modal, ParaView UI.
- **TCAA** — aeroacoustics: FW–H / Farassat 1A far-field, SPL/octaves, WAV.
- **TBASE** — SQLite simulation DB: reproducible archive, ParaView IPC, surrogates.
- **TMESH** — FEA volume meshing from STL/STEP (NetGen, Gmsh); assemblies.
- **TCFD / TOPT** — OpenFOAM orchestration and solver code; DoE/DIRECT/EGO.
- **TCAE DevOps** — GitLab CI, CTest + headless Qt, Linux/Windows installers.

2019–2020 **Research (doctoral studies)**, *Charles University (MFF UK)*, Prague, Computational biophysics / membrane modelling
Research in the group of Christoph Allolio on continuum modelling of lipid membranes.
- Ran molecular dynamics simulations (GROMACS) of lipid membranes.
- Implemented a continuum mechanics solver in C++ for membrane remodelling.

Education

2017–2020 **Master**, *Charles University*, Prague, Mathematical Modeling in Physics and Technology
Thesis: *Comparison of possible formulations of fluid-structure interactions with application in biomechanics*, supervised by RNDr. Jaroslav Hron, Ph.D.

2018–2019 **Erasmus**, *Technical University Munich*, Germany, Faculty of Mathematics
Machine learning, deep learning, high-performance computing.

2014–2017 **Bachelor**, *Charles University*, Prague, General Mathematics

Computer skills

Languages	C++ (C++23), Python, TypeScript, Java, Bash
Scientific	OpenFOAM, CalculiX, NetGen, Gmsh, ParaView, VTK, FEniCS, GROMACS
Platform	Qt, CMake, CTest, SQLite, GitLab CI, NSIS, Linux, Git
Web BE	Django, Spring Boot, FastAPI, GraphQL, PostgreSQL
Web FE	React, Svelte, Tailwind CSS, Vite
Robotics	ROS, Gazebo
AI / ML	RAG / LLM integration; surrogate modelling over parametric design spaces

Languages

Czech (native), English (C1), German (B1)

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